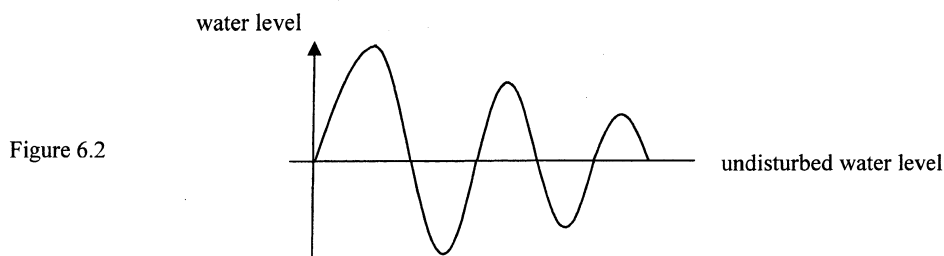


- (b) Figure 6.2 shows the profile of the water level along line L at a certain instant. Sketch on the same figure the profile at a time $\frac{T}{2}$ later, where T is the period of the water waves. (1 mark)



- (c) Q is a point on the line joining S_1 and S_2 as shown in Figure 6.1. State the kind of interference that occurs at Q and give a reason for this occurrence. (2 marks)

- *(d) A similar double-slit set-up is used for the demonstration of the interference of light in which the separation between slits S_1 and S_2 is 0.5 mm and the screen is at 2.5 m from the slits. Calculate the average separation between adjacent bright fringes on the screen for a monochromatic light of wavelength 550 nm. (2 marks)

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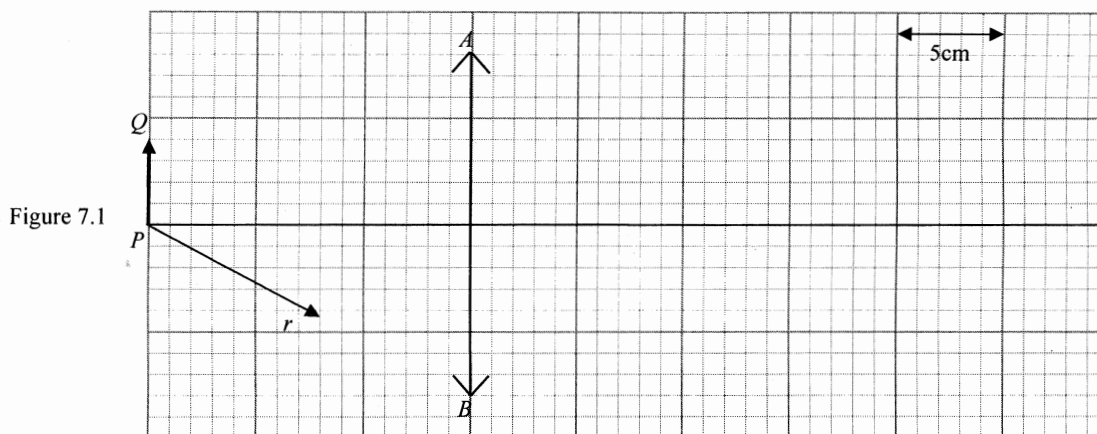
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7. A luminous object PQ is placed 15 cm in front of a convex lens AB as shown in Figure 7.1.

(a) The focal length of the lens is 5 cm.

- (i) Use a graphical method to find the location of the image of the object. Clearly draw all the construction lines on Figure 7.1 and state the nature of the image. (4 marks)



- (ii) Complete the path of ray r on Figure 7.1 to show how it travels after passing through the convex lens. (1 mark)

(b) Suppose that a convex lens of focal length 10 cm is used instead while the size of the lens and the object distance of PQ from the lens remain unchanged.

- * (i) Use the lens formula to find the image distance. Find also the linear magnification of the image. (3 marks)

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(ii) Compare the brightness of this image with that in (a). Explain.

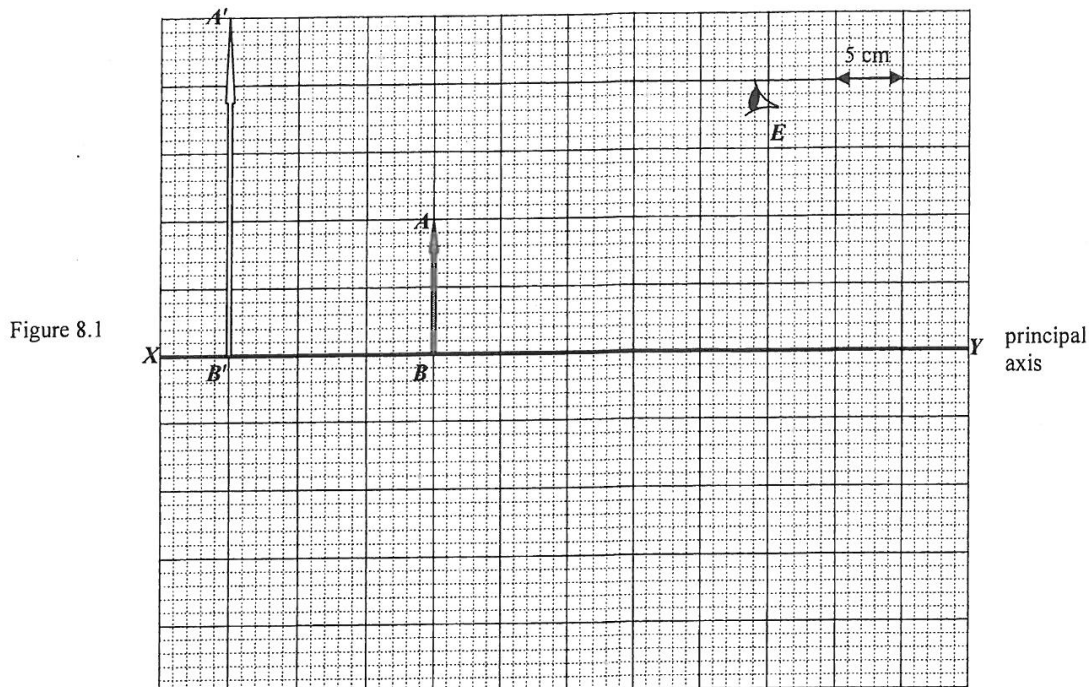
(2 marks)

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8. In Figure 8.1, $A'B'$ represents the image of an object AB formed by a lens L (not shown) where XY is the principal axis of the lens.



- (a) (i) Is the image real or virtual ? (1 mark)

- (ii) What kind of lens is used ? Explain your answer. (2 marks)

- (b) (i) Locate the optical centre O of lens L and draw on Figure 8.1 the position of lens L . (1 mark)

- (ii) By drawing an additional light ray, mark the principal focus F of the lens and find its focal length. The horizontal scale is 1 cm to 5 cm. (2 marks)

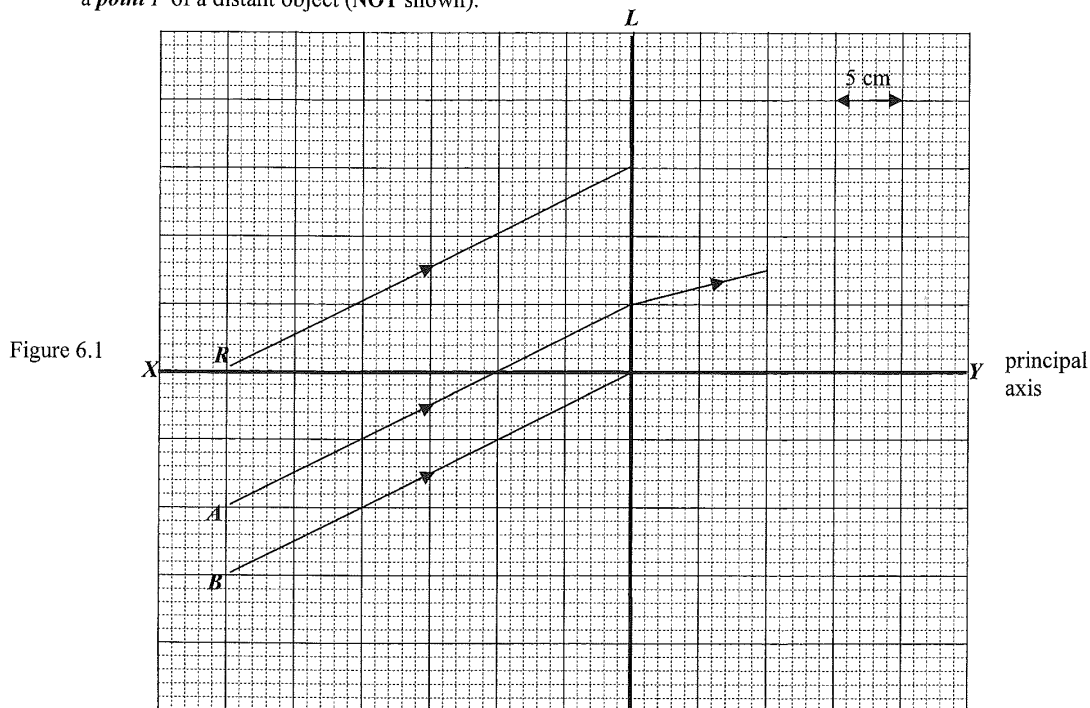
Focal length =

- (c) Draw a light ray to show how the eye E shown can see the image of head A through lens L . (2 marks)

- (d) State an application of lens L in the situation as shown above. (1 mark)

Answers written in the margins will not be marked.

6. In Figure 6.1, XY is the principal axis of a thin spherical lens L while A, B are two parallel rays coming from a point P of a distant object (NOT shown).



- (a) What kind of lens is L ? Explain. (2 marks)

- (b) (i) Locate the image of P (denoted it as point P'). (2 marks)

- (ii) Hence, determine the focal length of the lens. (1 mark)

Focal length =

- (c) R is a ray coming from the same point P ; complete its path after passing through the lens. (1 mark)

- (d) Based on the situation shown in the ray diagram above, describe a simple experimental method to determine the focal length of lens L . (2 marks)

Answers written in the margins will not be marked.

6. Read the following description about a **mirage** and answer the questions that follow.

A mirage is often seen on highways during hot summers. Pools of water seem to cover the roadway far ahead. Distant objects appear to be reflected by the surface of the 'water'. The phenomenon is caused by the difference in refractive index between the hot air near the road surface and the cooler air above it. The refractive index of cool air is greater than that of hot air, but the differences are so small that the subsequent deviations of light rays are tiny. Sufficiently large temperature differences between the hot air near the road surface and the above cooler air over a short height (i.e. high temperature gradient) and light rays travelling along sufficiently long path lengths are required to form a mirage.



Figure 6.1 Mirage seen on a highway. This photo was taken with a telephoto lens which gives the perception that the viewer is very close to the car ahead.

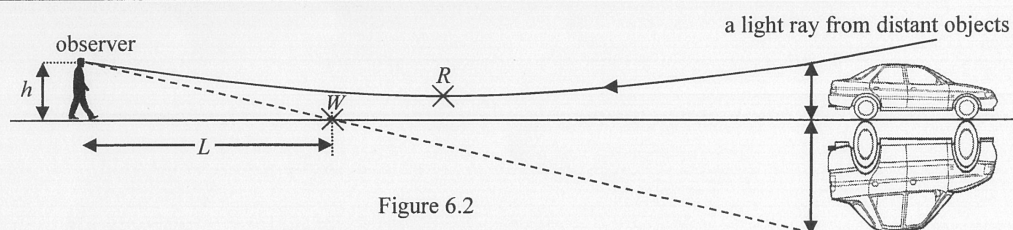


Figure 6.2

Figures 6.2 and 6.3 illustrate the principle of the phenomenon. Air of different temperatures is simplified to several layers and modeled as parallel slabs as shown in Figure 6.3. The bending of the light ray from distant objects is much exaggerated. θ_1 , θ_2 , θ_3 and θ_4 denote the angles of incidence at various boundaries of air layers.

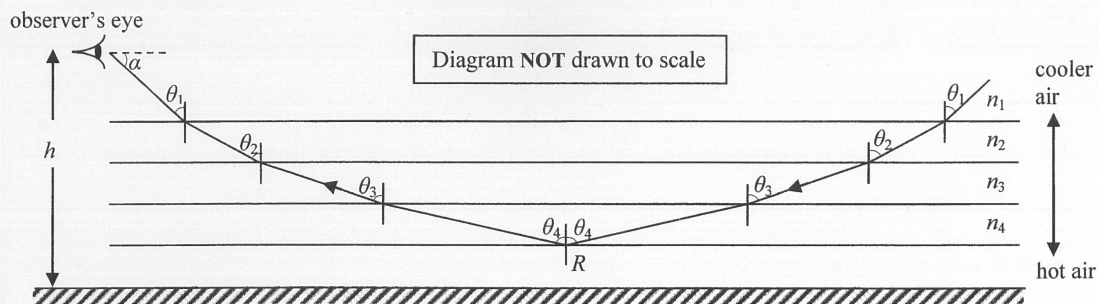


Figure 6.3

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(a) State **ONE** essential condition for a mirage to be observed.

(1 mark)

(b) (i) Referring to Figure 6.3, deduce the relationship between θ_1 , θ_4 and refractive indices n_1 , n_4 . For total internal reflection just to occur at R , θ_4 can be taken as 90° . Hence, find the corresponding value of θ_1 if $n_1 = 1.000261$ and $n_4 = 1.000221$. (3 marks)

(ii) Find L in Figure 6.2 if $h = 1.5$ m. (Note: $\alpha + \theta_1 = 90^\circ$ in Figure 6.3.)

(2 marks)

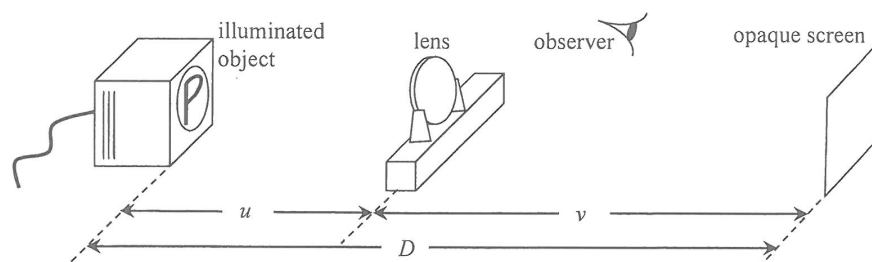
(c) A thirsty traveller in a vast desert sees similar mirages such that a 'water source' appears at W which is distance L away like the one in Figure 6.2. If he walks a distance L towards that 'water source', how far would the 'water source' appear to him? Explain your answer. (2 marks)

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5.

Figure 5.1



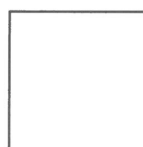
Kitty uses the set-up in Figure 5.1 to study the image formation of a lens. The lens is placed at a distance u from an illuminated object (letter 'P'). An opaque screen is placed at a distance D from the object so as to capture the image.

- (a) (i) State the kind of lens used. Explain your answer.

(2 marks)

- (ii) Sketch the image on the screen seen by the observer.

(1 mark)



opaque screen

- (b) The separation D is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance u is obtained for plotting a graph of D against u (Figure 5.2).

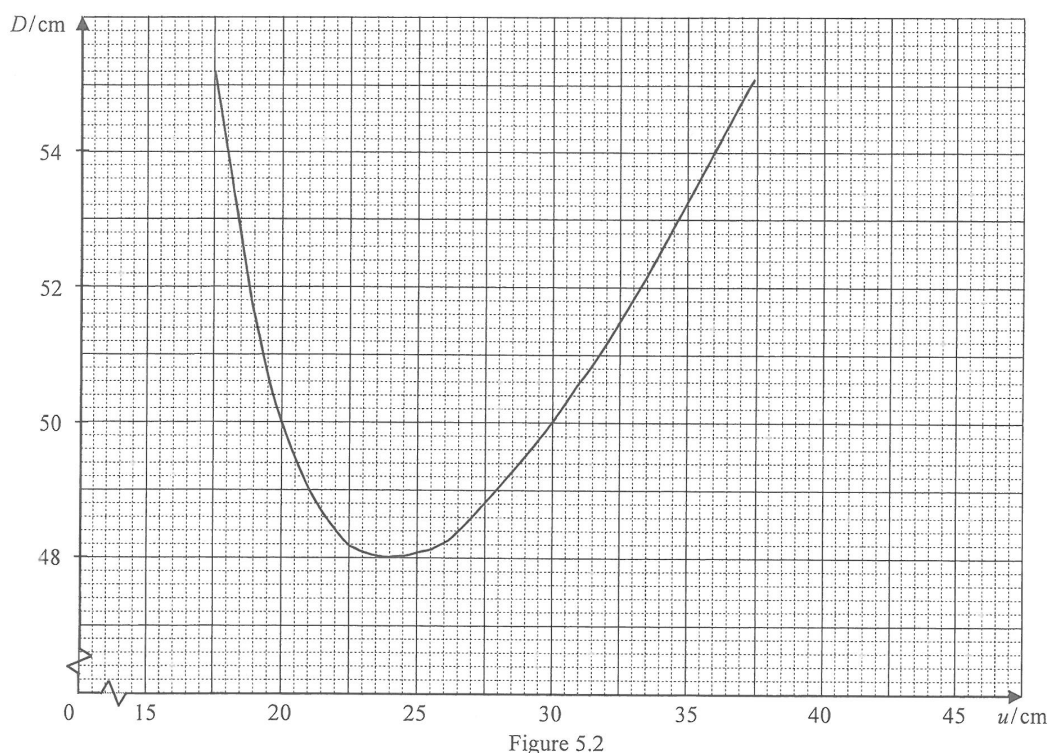


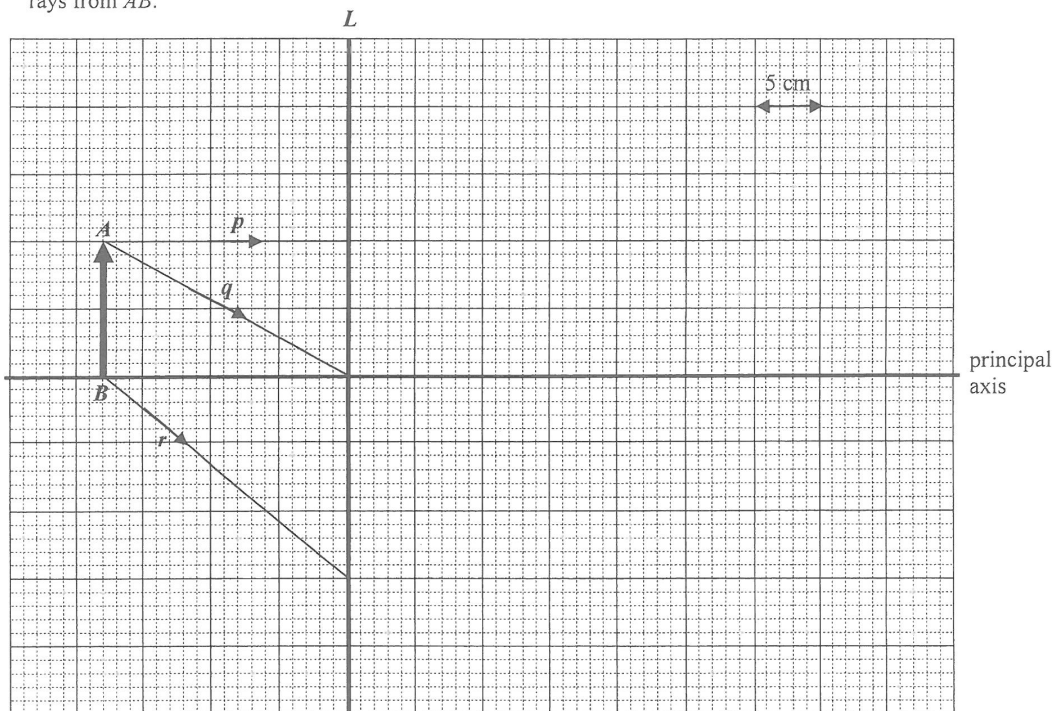
Figure 5.2

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- (i) When the lens is placed at 18 cm from the object, use the graph to find the corresponding separation between the lens and the screen. Hence calculate the magnification of the image. (2 marks)

In the figure below, AB represents the illuminated object which is at 18 cm from the lens L . p , q and r are light rays from AB .



- (ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of p , q and r . (3 marks)
- (iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

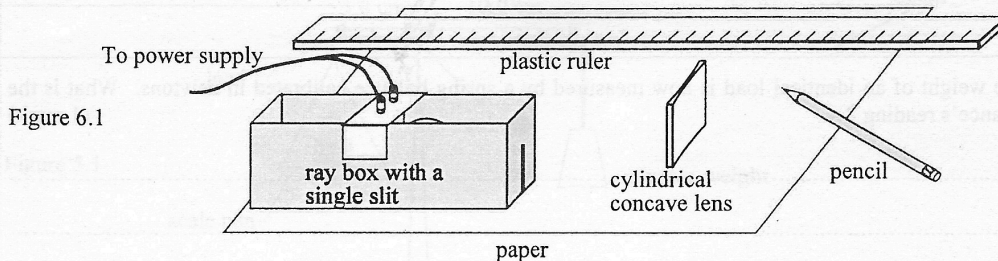
Focal length =

- (iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

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6. (a) You are given a ray box with a single slit (producing a fine light beam), a cylindrical concave lens, a plastic ruler, a pencil and a piece of paper as shown in Figure 6.1.



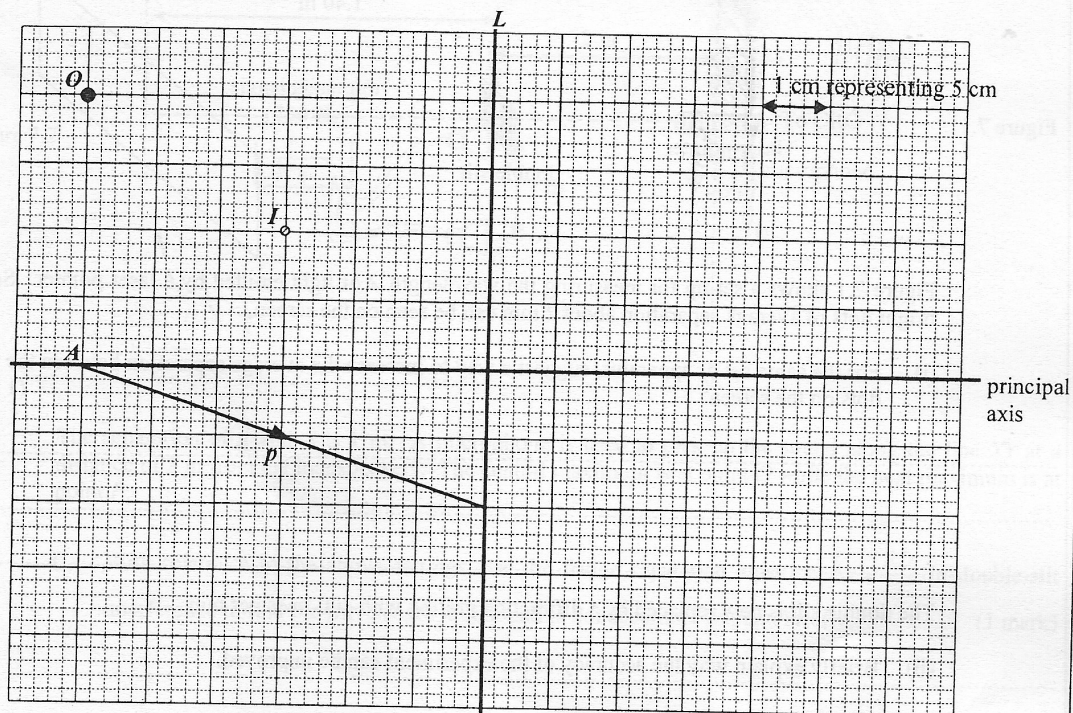
Describe how you would use the above apparatus to find the focal length of the lens and state **ONE** possible source of error in the experiment. (5 marks)

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- (b) In the figure below, L represents another cylindrical lens. A vertical pin used as the object is placed at O , the image is formed at I by the lens. The horizontal scale is 1 cm to 5 cm.



- (i) What kind of lens is used ? Explain.

(2 marks)

- (ii) Draw a suitable light ray to locate the principal focus F of lens L . Find its focal length.

Focal length =

(2 marks)

- (iii) Complete the path for the ray p from point A .

(1 mark)

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6. In Figure 6.1, AB represents the virtual image of an object formed by lens L . The magnification of the image is 0.4. The horizontal scale is 1 cm to 5 cm.

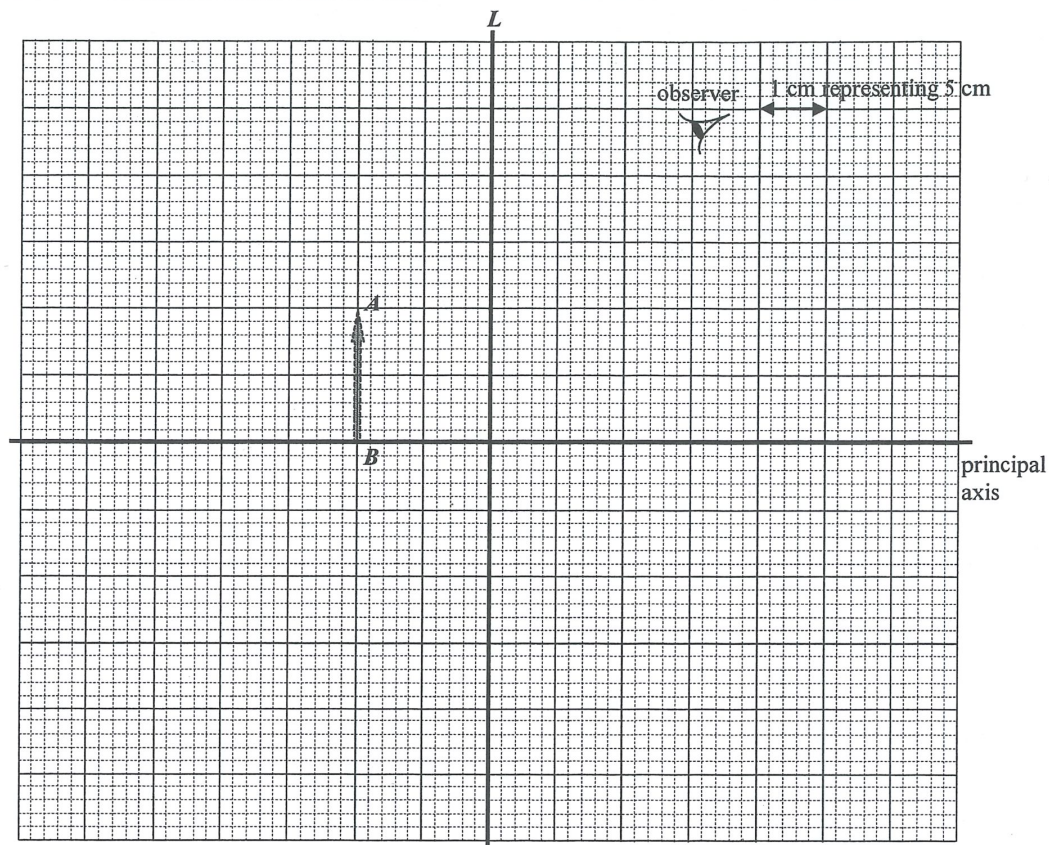


Figure 6.1

- (a) What kind of lens is used? Explain. (2 marks)

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- (b) Indicate on Figure 6.1 the position and height of the object. (2 marks)

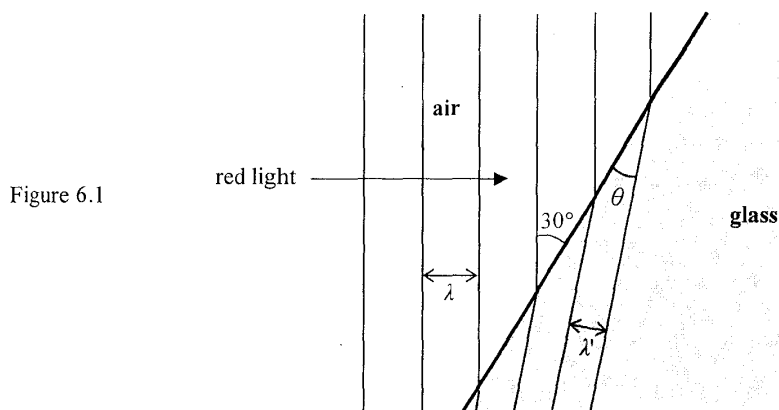
- (c) By drawing a suitable light ray, locate and mark the position of the focus, F , of the lens. Find the focal length of the lens. (3 marks)

Focal length =

- (d) Draw a light ray emerging from the object to illustrate how the observer in the figure can see the tip A of the image. (2 marks)

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6. (a) When red light of wavelength $\lambda = 675 \text{ nm}$ is incident at an angle of 30° from air to glass as shown in Figure 6.1, refraction occurs such that its wavelength in glass becomes $\lambda' = 450 \text{ nm}$ while the angle of refraction is θ .



- (i) What is the frequency of the red light in glass? (2 marks)

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- (ii) Find θ . (2 marks)

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- (iii) If the red light is replaced by blue light, θ will decrease. Compare the refractive index of glass for the red light and blue light. (1 mark)

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6. (b) Figure 6.2 shows a system for projecting a transparent photographic slide O onto a screen. The slide and the screen are 1 m apart. A beam of white light illuminates the slide. The position of lens L is adjusted until a sharp image of O magnified linearly to 9 times is formed on the screen.

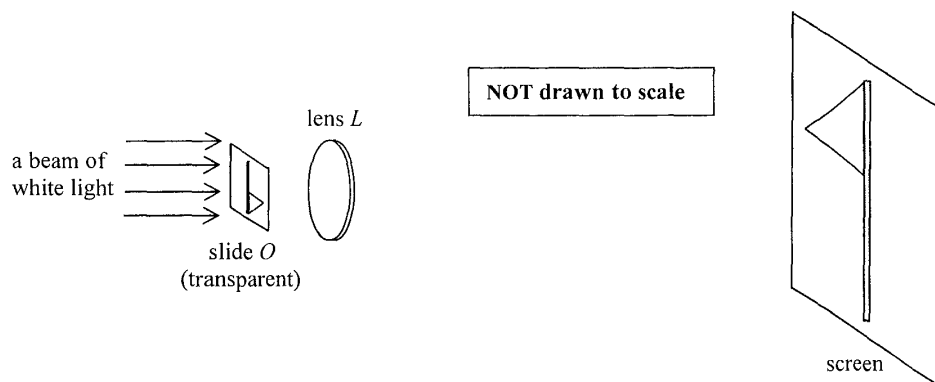
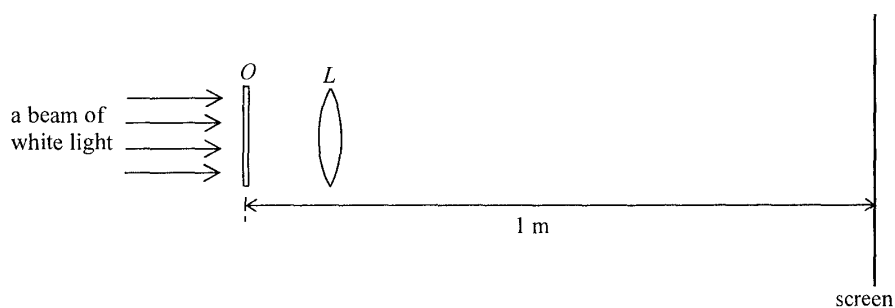


Figure 6.2



- (i) State the nature of this magnified image. (1 mark)

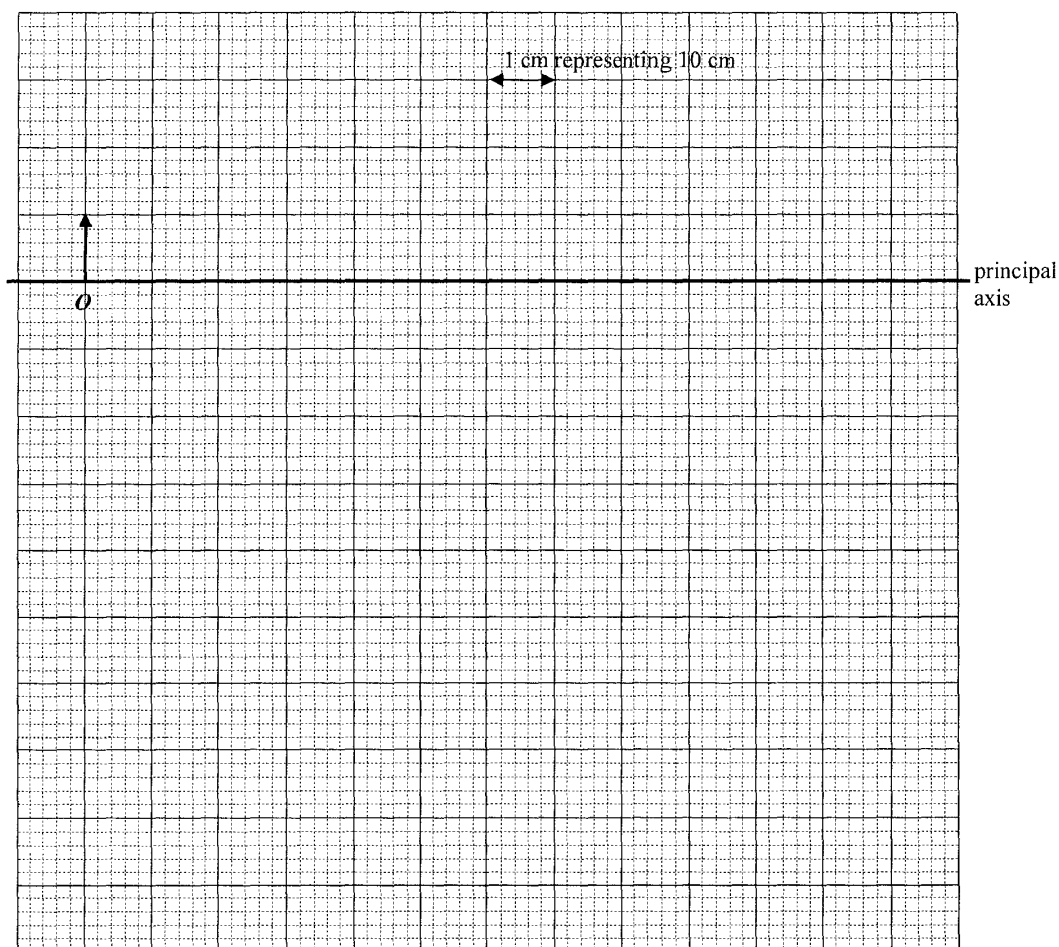
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- (ii) Find the separation between O and L . (1 mark)

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- (iii) Draw a ray diagram to show how the image of slide O is formed on the screen. Locate the focus F of lens L on your diagram and find the focal length of L . (Slide O and the principal axis of the lens have been drawn for you.) (5 marks)

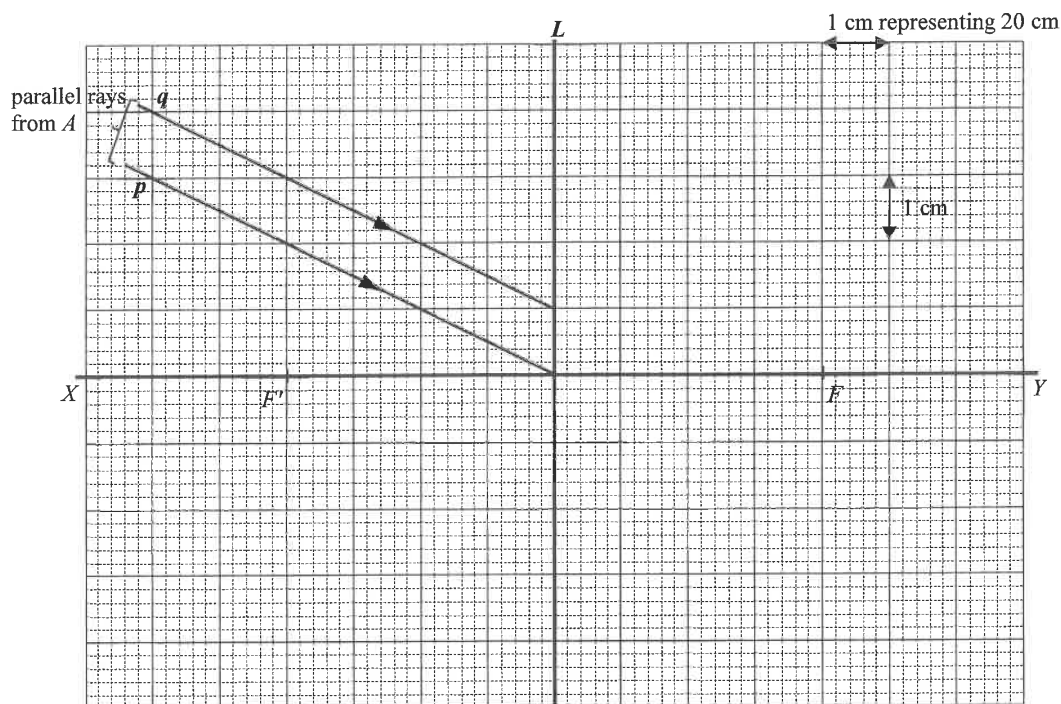


Focal length of L =

- (iv) When a black-and-white slide is projected onto the screen, the image has colour edges. Explain briefly. (Hint: the lens is made of glass.) (2 marks)

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5. In the figure below, XY is the horizontal principal axis of a convex lens L with principal foci F and F' . p, q are parallel rays coming from point A of a distant object AB . (The object can be represented by a vertical arrow $\begin{smallmatrix} A \\ \uparrow \\ B \end{smallmatrix}$ but it is **not** shown on the figure and its end B is on the principal axis).



- (a) (i) Draw the refracted rays of p and q so as to locate the image of A (denoted as A'). Hence mark the image $A'B'$ of object AB . (3 marks)
- (ii) Suggest an experiment to verify whether a real image is formed in the above situation. (2 marks)

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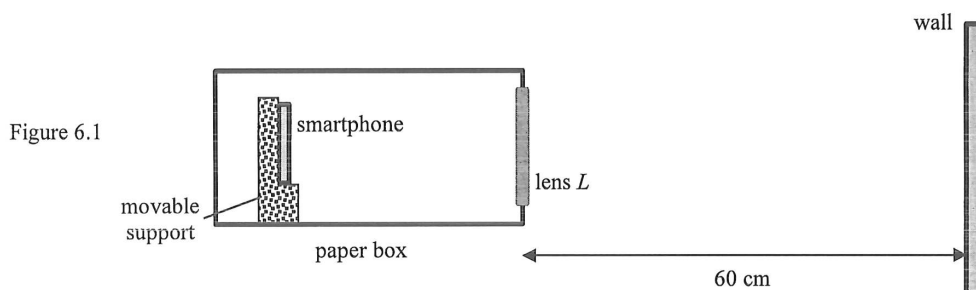
- (b) (i) Use the ray diagram drawn to estimate the ratio $\frac{\text{height of object } AB}{\text{distance of } AB \text{ from } L}$. The horizontal and vertical scales are 1:20 and 1:1 respectively. (2 marks)

- (ii) Hence, estimate the height of object AB which is a lamp post at a distance of 200 m from lens L . (1 mark)

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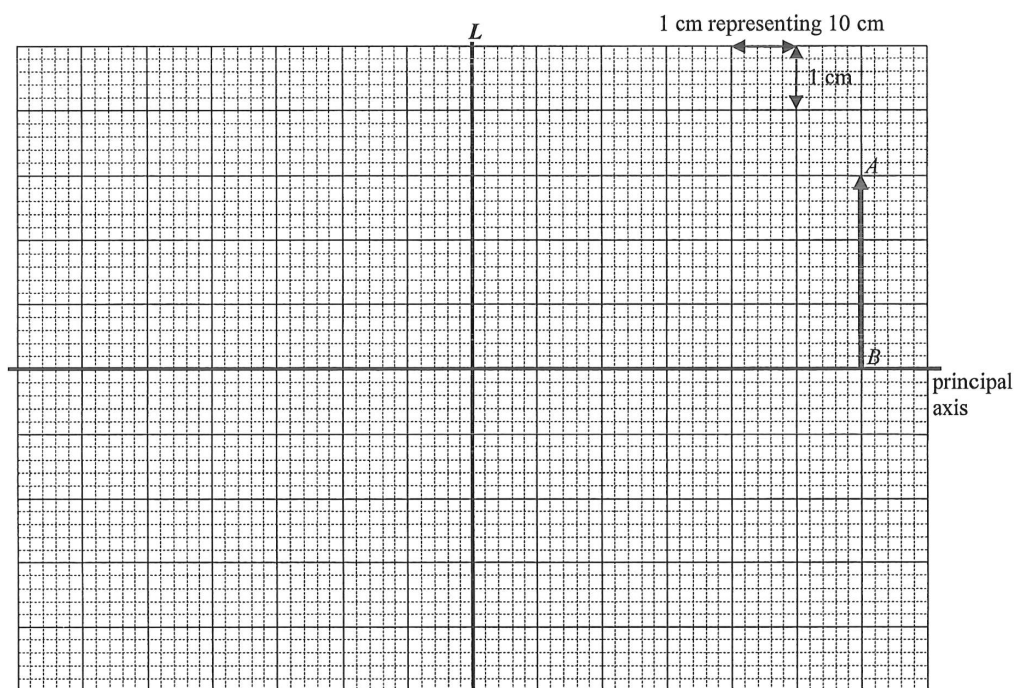
6. A student builds a smartphone projector as shown in Figure 6.1. It consists of a paper box with a lens L in front and a movable smartphone support inside. The position of the smartphone can be adjusted until a sharp image is formed on a wall.



- (a) What kind of lens is used? Explain briefly.

(2 marks)

- (b) In the figure below, AB represents the image formed on the wall, which is 60 cm from lens L . Given that the magnification of the image is 3.



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- (i) Locate and draw the object (i.e. the display on the smartphone) using an arrow and denote it as O . (1 mark)
- (ii) Complete the ray diagram by drawing **TWO** light rays reaching A from the object. (2 marks)
- (iii) Hence, or otherwise, find the focal length of L . (1 mark)

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- (c) If the separation between lens L and the wall is slightly increased, state the adjustment to be made so as to restore a sharp image on the wall. (1 mark)

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- (d) Suggest a way to minimise unwanted reflections inside the box. (1 mark)

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